

Exploring CNNs Through Image Classification

Raul Sebastien B. Baldemoro, Jan Vincent G. Elleazar

*Department of Computer Science, College of Information and Computing Sciences, University of Santo Tomas
España Blvd, Sampaloc, Manila, 1008 Metro Manila, Philippines*

raulsebastian.baldemoro.cics@ust.edu.ph

janvincent.elleazar.cics@ust.edu.ph

I. INTRODUCTION

Image classification is one of the classic computer vision problems in the field of machine learning. For this laboratory exercise, hair types dataset will be explored to train a Convolutional Neural Network (CNN) in classifying hair types. CNN has the capacity to systematically extract increasingly sophisticated representations of the content within an image (*ML Practicum: Image Classification*, 2025). The dataset is composed of 3 classes; curly hair, straight hair, and wavy hair.

The objective of this laboratory exercise is to train a convolutional neural network (CNN) model to predict the type of hair based on the given hair type dataset. Multiple experiments were conducted for analysis of different data preprocessing and hyperparameter tuning techniques.

II. METHODOLOGY

This section discusses the detailed approach of using data analysis techniques and machine learning modelling techniques to implement image classification using CNN. This section is segmented into three (3) subsections; Data Preprocessing, Data Splitting, and Model Implementation.

A. Data Preprocessing

Images within the dataset undergo a straightforward validation process to check the types that can be used by the *TensorFlow* library. Image types such as "*bmp*", "*gif*", "*jpeg*", "*png*" are ones that are considered valid to undergo preprocessing and training, and any files that don't meet said requirements are to be removed. This ensures that all files are match accepted formats

B. Data Splitting

Valid image types within the dataset are first standardized into 64x64 pixels. The dataset is then split into training and validation data; with a validation split of 80/20 (0.2) and batch size of 32 for efficient training. Categorical outputs are used for label encoding

C. Model Implementation

The CNN model was designed using a sequential architecture tailored for multi-class image classification. It consisted of stacked convolutional layers interleaved with max-pooling operations, which extracted hierarchical spatial features while progressively reducing the dimensionality of the data. These feature maps are then flattened and passed through dense layers, ending with a softmax output layer that produces class probabilities.

Adam optimizer was employed due to its efficiency and adaptability in handling sparse gradients. The loss function used is sparse categorical cross-entropy, with the model trained across multiple epochs with both training and validation performance tracked continuously.

III. EXPERIMENTS

A total of five (5) experiments were conducted with each having their own sub-experiments. Diverse combinations of architecture modification and hyperparameter tuning of CNN were tested to see how those combinations affect the performance of CNN in classifying hair types. The following are the said experiments:

- Experiment #1: Other Preprocessing methods for Images

- Experiment 1.1: Consistent Image Dimensions
- Experiment 1.2: Pixel Intensity Normalization
- Experiment 1.3: Filters and Edge Sharpening
- Experiment #2: Implementation of Different Types of Pooling
 - Experiment 2.1: Max Pooling
 - Experiment 2.2: Average Pooling
 - Experiment 2.3: Global Max Pooling
 - Experiment 2.4: Global Average Pooling
- Experiment #3: Changing Number of Filters
 - Experiment 3.1: filters 2, 4, 8
 - Experiment 3.2: filters 4, 6, 8
 - Experiment 3.3: filters 8, 16, 32
 - Experiment 3.4: filters 16, 32, 64
 - Experiment 3.5: filters 32, 64, 128
 - Experiment 3.6: filters 16, 16, 16
- Experiment #4: Changing Kernel Size
 - Experiment 4.1: Kernel Size (3,3), (5,5), (9,9)
 - Experiment 4.2: Kernel Size (3,3), (3,3), (3,3)
 - Experiment 4.3: Kernel Size (5,5), (5,5), (5,5)
 - Experiment 4.4: Kernel Size (7,7), (5,5), (3,3)
- Experiment #5: Changing Learning Rate
 - Experiment 5.1 Learning Rate 1e-2
 - Experiment 5.2 Learning Rate 1e-4
 - Experiment 5.3 Learning Rate 1e-5

Experiment #1 explores other data preprocessing techniques to improve image quality. The first sub-experiment aimed for all images in the dataset to have consistent image dimensions. Therefore, OpenCV was utilized to resize the width of all images into 224px while maintaining their respective aspect ratio to prevent image distortion. The second sub-experiment aimed for the pixel intensity of all images to be normalized. This was done by utilizing NumPy to convert the image data into float32 and then dividing by 255.0 for the pixel intensities will be scaled to the range [0,1]. The last sub-experiment aimed to apply filtering and edge sharpening techniques to all images. This was done

by utilizing OpenCV to apply Gaussian Blur and Laplacian Filter to all images. Gaussian Blur is used for image smoothing and noise reduction while Laplacian Filter is used for edge sharpening.

Experiment #2 explores the implementation of different types of pooling. Pooling layer is used in CNNs to reduce the spatial dimensions (width and height) of the input feature maps while retaining the most important information (GeeksforGeeks, 2025). The first sub-experiment used Max Pooling by utilizing the MaxPooling2D function of Keras. A Max Pooling layer was added after each Conv2D layer. The second sub-experiment used Average Pooling by utilizing the AveragePooling2D of Keras. An Average Pooling layer was added after each Conv2D layer. The third sub-experiment used Global Max Pooling by utilizing the GlobalMaxPooling2D of Keras. A Global Max Pooling layer was added after the last Conv2D layer. The fourth sub-experiment used Global Average Pooling by utilizing the GlobalAveragePooling2D of Keras. A Global Average Pooling layer was added after the last Conv2D layer.

Experiment #3 explores the effects of different numbers of filters. The first two sub-experiment used lower numbers of filters based on the baseline number of filters. The next three sub-experiments used higher numbers of filters based on the baseline number of filters. The last sub-experiment used the same number of filters which is 16.

Experiment #4 explores the effects of different kernel sizes. The first sub-experiment used increasing kernel size compared to the baseline. The next two sub-experiments used constant kernel sizes namely (3,3), (3,3), (3,3), and (5,5), (5,5), (5,5). The last sub-experiment used decreasing kernel size compared to the baseline.

Lastly, Experiment #5 explores the effects of different learning rates of the model. The first sub-experiment used a learning rate of 1e-2. The second experiment used a learning rate of 1e-4. The last experiment used a learning rate of 1e-5.

IV. RESULTS AND ANALYSIS

A. Experiment #1: Other Preprocessing Methods for Images

1. CONSISTENT IMAGE DIMENSIONS:

In this sub-experiment, OpenCV library was utilized to resize the width of all images in the dataset to 224 pixels and keep its aspect ratio. Cropping was not used intentionally to preserve the full content of each hair sample. The intention was to standardize the visual structure of the dataset and, hopefully, improve feature extraction when training.

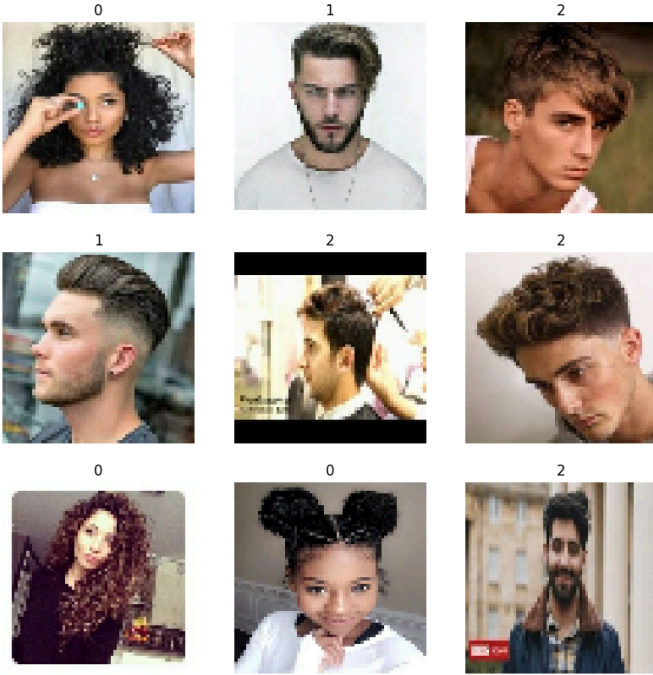


Fig. 1. Sample Visualization of Preprocessed Dataset for Experiment 1.1

The CNN model was trained over 50 epochs using the resized dataset. The performance progression during training showed consistent learning behavior, with the training accuracy gradually increasing and stabilizing around 55.73% by the final epoch. Meanwhile, the validation accuracy followed a similar upward trend, reaching 52.55% at epoch 50.

Loss values for both training and validation also exhibited a smooth downward trajectory, with final values converging near 0.91 and 0.99 respectively, illustrated in Fig. 2. The absence of sharp divergences between the training and validation curves suggests that the model generalized

reasonably well, and no major overfitting was observed. While the overall accuracy remained moderate, the consistent convergence indicates that the resized, standardized images contributed to a more stable and learnable dataset structure.

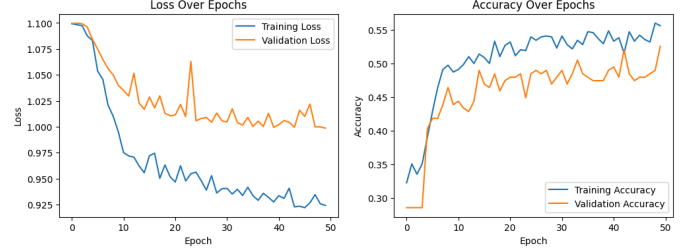


Fig. 2. Accuracy and Loss Functions over Epochs for Experiment 1.1

To evaluate how well the model distinguished between hair types, we analyzed the classification metrics on the validation set. The results, summarized in Table I, indicate that the model performed best in identifying curly hair, with an F1-score of 0.56, while straight hair presented more classification challenges.

TABLE I
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 1.1

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.62	0.52	0.56
STRAIGHT HAIR	0.49	0.51	0.50
WAVY HAIR	0.51	0.55	0.53

The confusion matrix in Fig. 3 presents the detailed breakdown of the model's predictions across the three hair type classes. The diagonal values represent correct classifications: 29 for curly hair, 34 for straight hair, and 40 for wavy hair. Misclassifications are found off-diagonal, notably between curly and wavy hair, and between wavy and straight hair.

This confusion pattern suggests that wavy hair, in particular, tends to be mistaken for straight hair (26 instances), likely due to their similar texture under certain lighting or angles. Conversely, straight hair was classified with the least ambiguity, showing

only 11 and 22 misclassifications to curly and wavy, respectively.

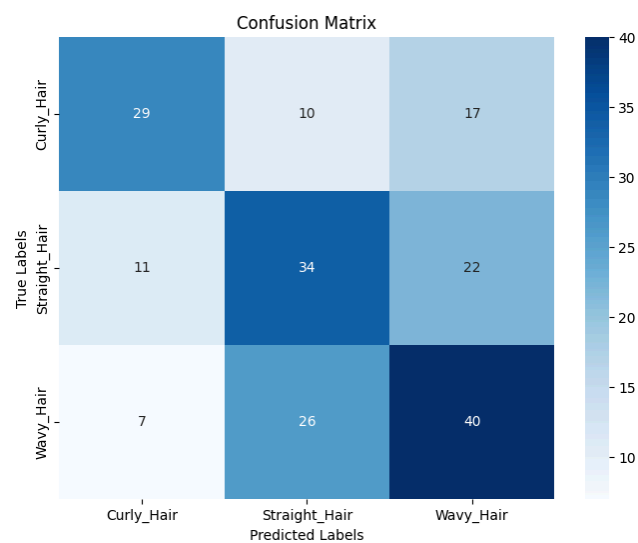


Fig. 3. Confusion Matrix for Experiment 1.1

2. PIXEL INTENSITY NORMALIZATION

In the second sub-experiment, the dataset images were further preprocessed through pixel intensity normalization. This preprocessing step converted image pixel values into floating-point representations and scaled them to a standardized range from 0 to 1. Normalization was performed by dividing pixel intensity values by 255.0, aiming to enhance the CNN’s training efficiency by improving numerical stability and potentially facilitating faster convergence.

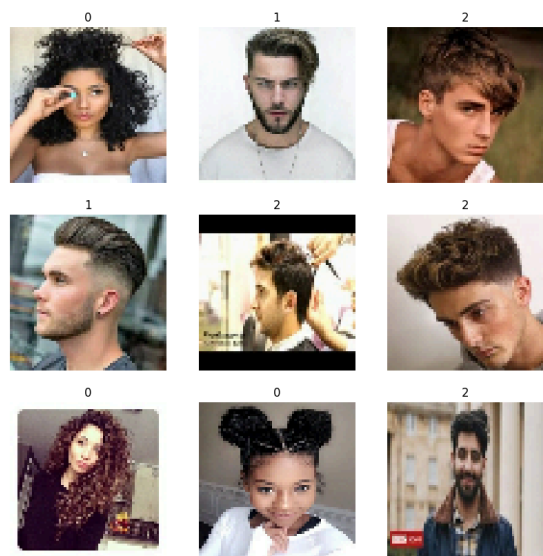


Fig. 4. Sample Visualization of Preprocessed Dataset for Experiment 1.2

After normalization, the CNN model was retrained over 50 epochs. As shown in Fig. 5, the training accuracy exhibited a smoother progression than the previous sub-experiment, achieving 60.2% by the final epoch. Validation accuracy improved moderately, stabilizing around 57.8% at epoch 50.

Correspondingly, the loss curves displayed steady declines, reaching 0.87 for training loss and approximately 0.95 for validation loss. These observations indicated that pixel intensity normalization positively impacted learning stability and model convergence.

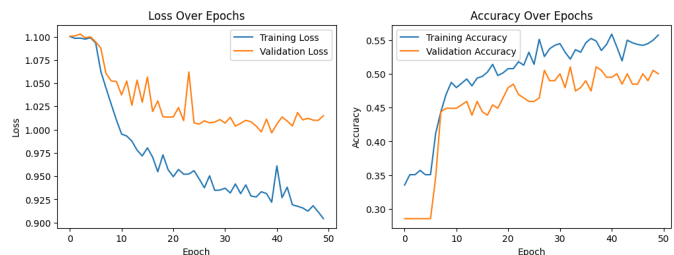


Fig. 5. Accuracy and Loss Functions over Epochs for Experiment 1.2

The classification report summarized in Table II shows decreased slightly compared to consistent image dimension preprocessing alone. Specifically, notable performance across all classes remained within a close margin of variation. Straight hair classification remained roughly consistent, though not significantly improved.

TABLE 2
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 1.2

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.58	0.45	0.51
STRAIGHT HAIR	0.47	0.51	0.49
WAVY HAIR	0.49	0.53	0.51

Further analysis via the confusion matrix (Fig. 6) reveals similar misclassification trends to Experiment 1.1. While straight hair was still classified with moderate reliability, confusion between wavy and straight hair remained high. These results suggest that while pixel intensity normalization maintained stable performance, it did not yield significant improvements over the prior preprocessing technique.

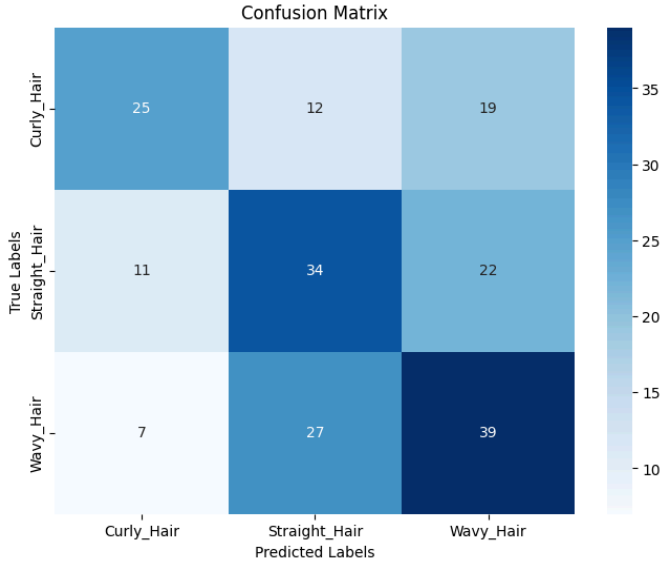


Fig 6. Confusion Matrix for Experiment 1.2

Pixel intensity normalization produced performance metrics that were largely comparable to those in Experiment 1.1. While the normalization step provided a numerically stable input range and may have contributed to smoother learning dynamics, it did not translate into a clear improvement in classification accuracy or F1-scores. The confusion matrix indicates that visual ambiguity particularly between wavy and straight hair types persisted even after normalization. These results suggest that normalization alone is not a strong enough signal-enhancing strategy for this classification task. More advanced preprocessing techniques, such as edge detection or contrast enhancement, may be required to make finer distinctions between similar hair textures. These approaches will be explored in the subsequent experiment.

3. FILTERS AND EDGE SHARPENING:

In this sub-experiment, additional image enhancement techniques were applied to emphasize texture features that could aid the CNN in distinguishing between similar hair types. Two filters were employed: Gaussian Blur and the Laplacian filter. Gaussian Blur was used first to reduce image noise and smooth out fine-grain inconsistencies, followed by the Laplacian filter to sharpen edges and enhance boundaries. This preprocessing aimed to strengthen the contrast of

hair patterns and emphasize structural differences between curly, straight, and wavy hair.

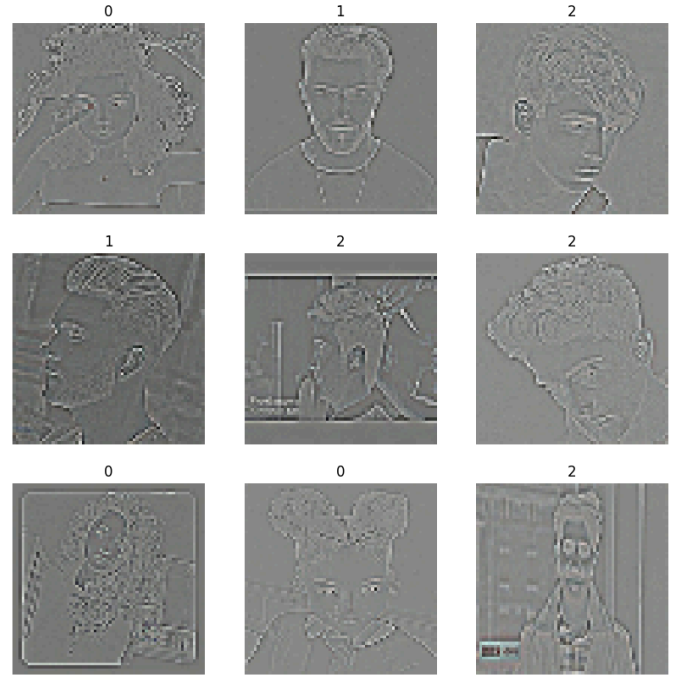


Fig. 7. Sample Visualization of Preprocessed Dataset for Experiment 1.3

As shown in Fig. 8, the CNN model trained on filtered and sharpened images exhibited poor convergence. By epoch 50, training accuracy reached only 34.80%, while validation accuracy plateaued at 28.57%, the lowest performance across all sub-experiments thus far. Loss values remained above 1.09, indicating that the model struggled to learn discriminative patterns effectively.



Fig. 8. Accuracy and Loss Functions over Epochs for Experiment 1.3

These results suggest that the combined use of Gaussian blur and Laplacian edge detection introduced noise or suppressed essential texture features that are critical for distinguishing hair types.

TABLE 3
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 1.3

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.29	1.00	0.44
STRAIGHT HAIR	0.00	0.00	0.00
WAVY HAIR	0.00	0.00	0.00

The classification report in Table III confirms that the model only successfully predicted the ‘‘Curly Hair’’ class. Both ‘‘Straight Hair’’ and ‘‘Wavy Hair’’ were completely unrecognized, resulting in zero precision, recall, and F1-score for those classes. This indicates a significant model collapse in class discrimination.

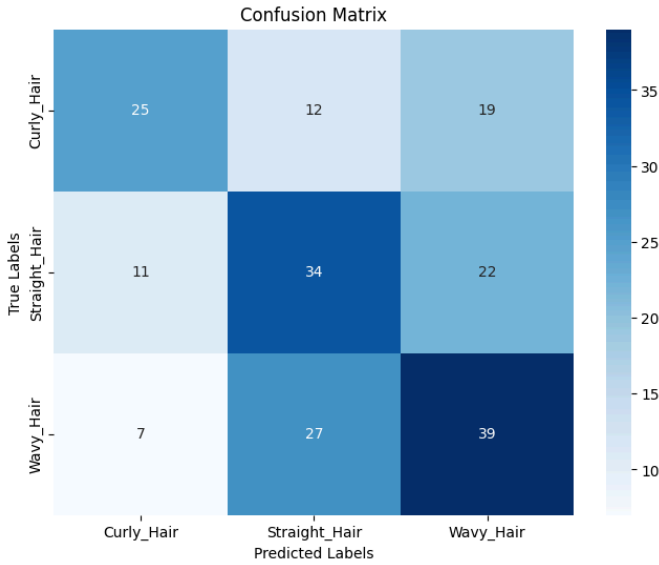


Fig 9. Confusion Matrix for Experiment 1.3

The confusion matrix (Fig. 9) further illustrates the imbalance in predictions. While actual curly hair samples were correctly identified, a significant number of straight and wavy hair images were incorrectly classified as curly, resulting in extremely poor class separation.

This experiment demonstrated that applying Gaussian Blur and Laplacian edge sharpening significantly degraded the model's performance. The CNN exhibited collapsed classification behavior, favoring a single class and failing to learn meaningful distinctions between hair types. This suggests that excessive filtering can suppress important mid-frequency features, disrupting the

balance of patterns necessary for effective learning. Filtering-based preprocessing should therefore be approached with caution, especially for fine-grained visual classification tasks such as hair type detection.

B. Experiment #2: Implementation of Different Types of Pooling

1. MAX POOLING

This sub-experiment tested the effect of using Max Pooling in the CNN architecture. MaxPooling2D layers were inserted after each Conv2D layer. The goal was to retain the most prominent feature activations while reducing spatial dimensionality and model complexity.

As shown in Fig. 8, the model trained for 50 epochs and reached a training accuracy of 78.26% and validation accuracy of 58.16%. Throughout training, both training and validation losses steadily decreased, with final values of 0.4889 and 1.1600 respectively. While the gap between training and validation loss widened in the later epochs, the model still showed signs of generalization.

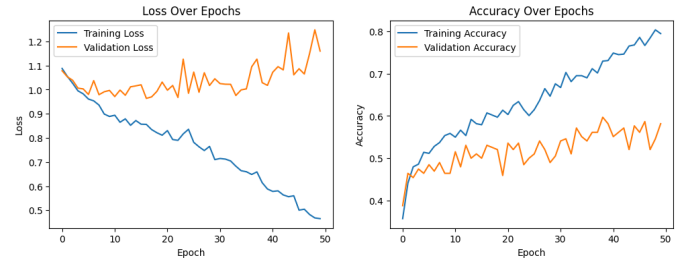


Fig. 9. Accuracy and Loss Functions over Epochs for Experiment 2.1

The classification report summarized in Table IV reflects balanced performance across all three classes. Curly hair achieved the highest F1-score of 0.65, while straight and wavy hair followed closely with 0.54 and 0.56 respectively. Precision and recall were also well distributed, indicating that the model was able to detect all classes without strong bias toward any particular category.

TABLE 4
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 2.1

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.60	0.71	0.65

STRAIGHT HAIR	0.57	0.51	0.54
WAVY HAIR	0.58	0.55	0.56

The confusion matrix in Fig. 9 supports the classification findings. The model correctly classified the majority of curly hair images (40/56), with moderate confusion between straight and wavy hair. Although some overlap remains between adjacent classes, particularly wavy-to-curly (15 errors) and wavy-to-straight (18 errors), the overall structure of predictions shows reasonable separation.

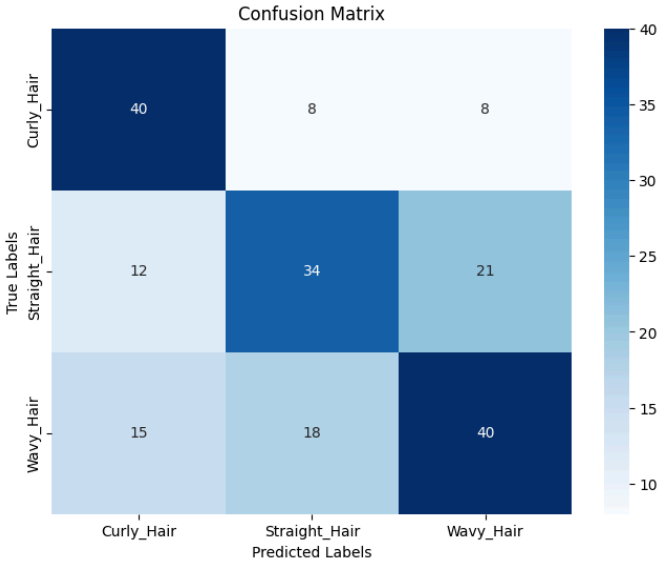


Fig 10. Confusion Matrix for Experiment 2.1

Max Pooling enabled the CNN to prioritize dominant spatial features and discard redundant information, which helped in achieving improved classification stability. The model demonstrated solid accuracy and balanced precision/recall across all classes, marking this as the best-performing configuration so far. However, some residual confusion—particularly between visually similar hair textures like straight and wavy—suggests that Max Pooling, while effective, may still lose subtle pattern details. Further experiments will evaluate alternative pooling methods such as average and global pooling to test whether they can improve upon this performance by retaining a more nuanced feature map.

2. AVERAGE POOLING

This sub-experiment implemented Average Pooling in the CNN architecture. AveragePooling2D layers were added after each convolutional layer to reduce dimensionality by computing the average of pixel values in local regions. This pooling method preserves a broader range of features but at a potentially reduced contrast compared to Max Pooling.

As shown in Fig. 11, the model trained over 50 epochs and reached a final training accuracy of 71.43% and validation accuracy of 56.12%. The training loss decreased consistently, ending at 0.6557, while the validation loss remained stable near 1.0579. Despite a slightly lower peak than Max Pooling, the training was stable and showed no significant overfitting.

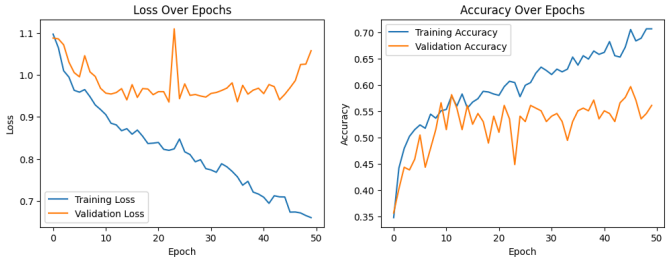


Fig. 11. Accuracy and Loss Functions over Epochs for Experiment 2.2

The classification report in Table V presents the performance across all classes. Curly hair retained the highest F1-score at 0.62, while straight hair trailed at 0.46, due to lower recall (0.37). Wavy hair showed moderate classification results with an F1-score of 0.58. The model's ability to identify straight hair samples was the weakest aspect of this configuration.

TABLE 5
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 2.2

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.58	0.68	0.62
STRAIGHT HAIR	0.61	0.37	0.46
WAVY HAIR	0.53	0.64	0.58

The confusion matrix in Fig. 12 further illustrates the class-wise distribution of predictions. Curly hair was classified most accurately, with 38 true positives and relatively few misclassifications.

However, straight hair saw high confusion, particularly with wavy hair (28 instances). Wavy hair classification showed better separation but still overlapped with both other classes.

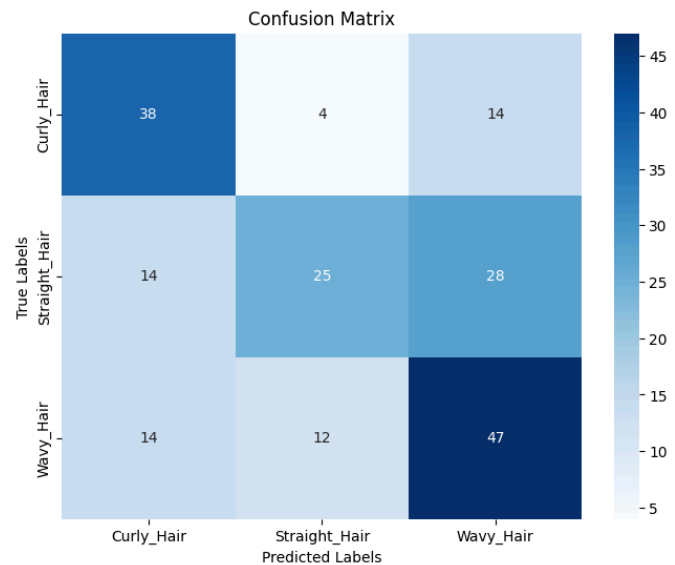


Fig 12. Confusion Matrix for Experiment 2.2

Average Pooling provided more general representations by retaining distributed feature values rather than focusing on extremes. While this helped in recognizing texture-rich classes like curly and wavy hair, it hindered the model’s ability to confidently classify straight hair, likely due to its subtler features. The performance was comparable to Max Pooling but slightly less balanced, with weaker recall in the straight hair class. In scenarios requiring finer detail retention, average pooling may underperform.

3. GLOBAL MAX POOLING

In this sub-experiment, Global Max Pooling (GMP) was employed to condense feature maps by retaining the single highest activation across the entire spatial dimension of each feature map. This technique dramatically reduces the number of parameters while preserving the most dominant features, aiming to improve generalization and reduce overfitting.

As shown in Fig. 13, the CNN reached a training accuracy of 79.30% and a validation accuracy of 62.76% by epoch 50 — the highest validation performance recorded thus far. The loss values also

declined to 0.4934 (training) and 0.9140 (validation), indicating stable and consistent convergence without significant overfitting.



Fig. 13. Accuracy and Loss Functions over Epochs for Experiment 2.3

As detailed in Table VI, curly and wavy hair classes saw substantial gains in both recall and F1-score, reaching 0.68 and 0.65 respectively. Straight hair, however, continued to pose challenges, with recall at 0.49 and F1-score at 0.55. The macro and weighted F1-scores both reached 0.63, the highest overall across all experiments to date.

TABLE 6
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 2.3

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.66	0.71	0.68
STRAIGHT HAIR	0.61	0.49	0.55
WAVY HAIR	0.62	0.68	0.65

The confusion matrix in Fig. 14 reinforces the improvements. Wavy hair classification reached 50 correct predictions, and curly hair achieved 40, while straight hair still faced modest confusion (13 mislabeled as curly, 21 as wavy). This suggests that GMP helped separate textural and edge-based differences effectively in more complex classes.

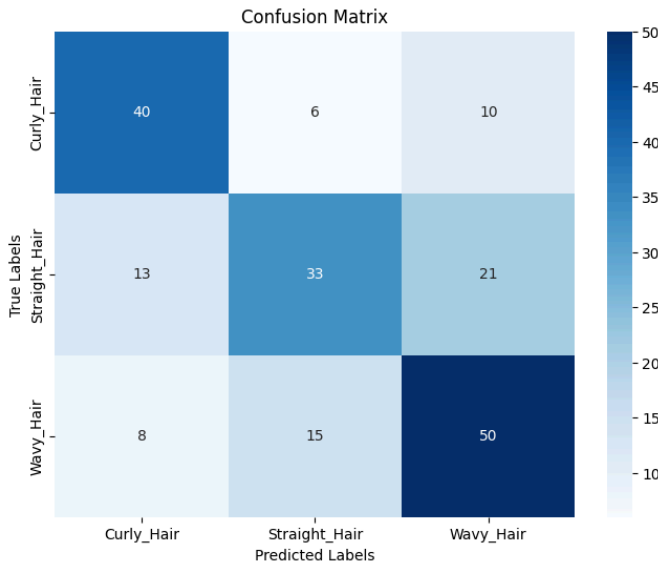


Fig 14. Confusion Matrix for Experiment 2.3

Global Max Pooling delivered the strongest overall performance among all pooling strategies tested. By condensing feature maps into the most salient activations, the model effectively emphasized dominant structural features—particularly in the curly and wavy hair classes—leading to high F1-scores of 0.68 and 0.65, respectively. However, straight hair remained more difficult to classify accurately, suggesting that subtle, low-frequency patterns were not fully captured by this pooling method.

While Global Max Pooling yielded the best overall classification results, the increasing gap between training and validation performance, along with a plateauing validation accuracy and loss, suggests mild overfitting. This indicates the model may have become overly reliant on dominant training features, limiting generalization beyond the training set.

4. GLOBAL AVERAGE POOLING

This sub-experiment tested Global Average Pooling (GAP), a technique that reduces each feature map to a single average value. GAP offers strong parameter reduction and spatial regularization by summarizing the entire feature map, potentially reducing overfitting and improving model interpretability.

As illustrated in Fig. 14, the model trained for 50 epochs, reaching a final training accuracy of 57.54% and validation accuracy of 49.49%. Unlike other pooling methods, validation loss remained relatively flat and noisy, ending at 1.0047, while training loss dropped to 0.8740. These signs, combined with early flattening of the validation accuracy curve, suggest the model struggled to generalize using GAP.

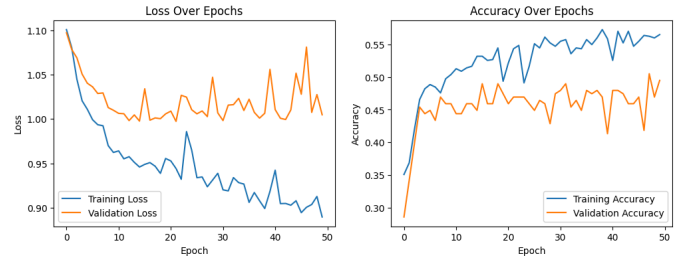


Fig. 15. Accuracy and Loss Functions over Epochs for Experiment 2.4

The classification report in Table VII highlights balanced yet low performance across all classes. Curly hair achieved the highest F1-score at 0.55, while wavy and straight hair lagged behind at 0.48 and 0.46 respectively. With macro and weighted F1-scores at 0.50, this configuration underperformed relative to all other pooling types.

TABLE 7
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 2.4

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.53	0.57	0.55
STRAIGHT HAIR	0.50	0.42	0.46
WAVY HAIR	0.50	0.51	0.48

The confusion matrix in Fig. 16 reflects broad prediction overlap between all classes. Although the model correctly identified many curly hair samples (32), misclassifications into wavy hair (17) were frequent. Straight and wavy hair were often confused with one another, particularly with 26 straight hair images misclassified as wavy and 21 wavy as straight.

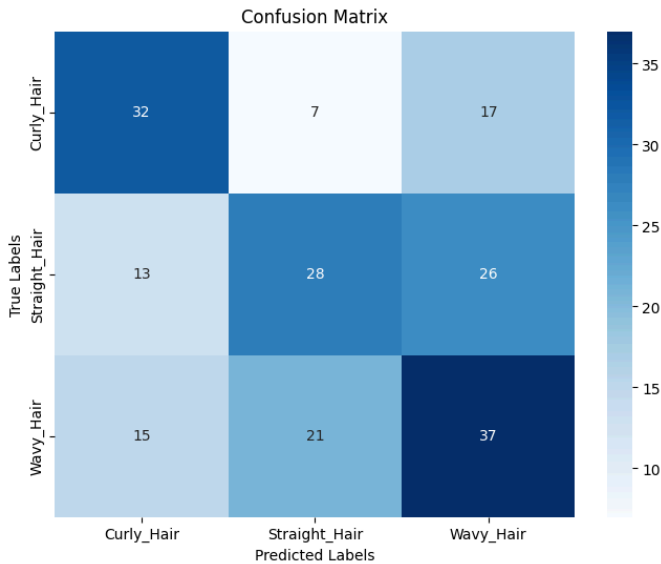


Fig 16. Confusion Matrix for Experiment 2.4

Global Average Pooling, while computationally efficient and regularizing, resulted in the lowest overall performance among all pooling methods evaluated. Its aggressive spatial averaging may have suppressed key local features necessary for distinguishing subtle hair texture variations. Although curly hair retained decent separability, straight and wavy hair predictions suffered significant overlap. These findings suggest that while GAP is useful in certain classification tasks, its effectiveness is limited for problems requiring fine-grained spatial discrimination.

C. Experiment #3: Changing Number of Filters

1. FILTERS 2, 4, 8

This sub-experiment investigated the effect of using a minimal convolutional filter configuration consisting of 2, 4, and 8 filters across three convolutional layers. The aim was to evaluate whether a lightweight model could still extract meaningful hierarchical features for hair type classification, and how model capacity constraints would affect learning and generalization.

As shown in Fig. 17, the model achieved a training accuracy of 64.19% and a validation accuracy of 46.94% after 50 epochs. Training loss steadily declined to 0.7911, whereas validation loss plateaued around 1.14, indicating limited generalization. The flattening of the validation accuracy and slight increase in validation loss

toward the end of training further suggest the model had difficulty fitting unseen data.

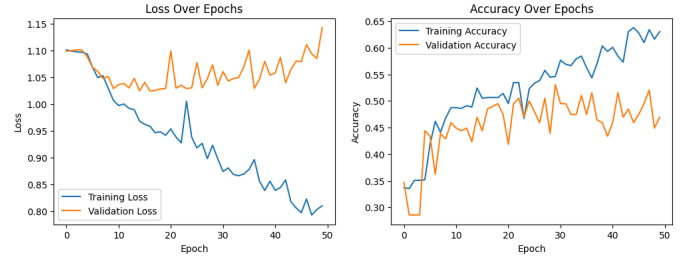


Fig. 17. Accuracy and Loss Functions over Epochs for Experiment 3.1

The classification report in Table VIII shows that curly hair was the most accurately identified class, with an F1-score of 0.60. Wavy hair followed with 0.51, while straight hair performance lagged, achieving only 0.29. The overall validation accuracy was 47%, indicating that this filter configuration lacked sufficient representational capacity for high-quality discrimination.

TABLE 8
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 3.1

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.58	0.62	0.60
STRAIGHT HAIR	0.33	0.25	0.29
WAVY HAIR	0.47	0.55	0.51

The confusion matrix in Fig. 18 highlights clear trends. While curly hair was reasonably well classified (35 correct), straight hair was frequently misclassified as wavy (34 errors). Wavy hair exhibited a moderate distribution, with some overlap into the curly class (11 errors). The imbalance indicates that the model lacked the depth to fully separate overlapping visual features across classes.

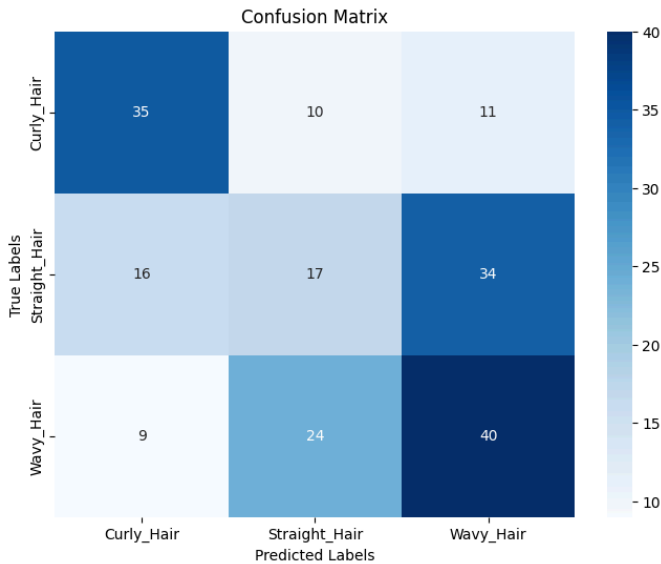


Fig 18. Confusion Matrix for Experiment 3.1

The use of a minimal filter configuration (2-4-8) constrained the model's ability to learn robust features, especially for less textured classes like straight hair. While curly and wavy hair benefited modestly, overall classification performance was weak. The results suggest that this shallow configuration does not provide sufficient hierarchical abstraction for complex classification tasks. Future sub-experiments will test deeper architectures with increasing filter counts to assess scalability and the trade-off between model complexity and performance.

2. FILTERS 4, 6, 8

This sub-experiment increased the number of filters per convolutional layer to 4, 6, and 8, respectively. The goal was to evaluate whether a modest increase in model capacity would yield improvements in feature extraction and classification performance without significantly increasing computational cost.

As seen in Fig. 19, the model exhibited a smoother training trajectory compared to the previous filter configuration. It achieved a training accuracy of 75.49% and a validation accuracy of 52.04%. Training loss declined to 0.5903, while validation loss stabilized around 1.06. Although validation performance plateaued early, it remained consistently above the previous sub-experiment.



Fig. 19. Accuracy and Loss Functions over Epochs for Experiment 3.2

The classification report in Table IX reveals improved balance across classes. Curly hair classification remained strong with an F1-score of 0.61, while wavy hair saw moderate success at 0.55. Straight hair classification remained weaker, though it improved slightly from the prior setup, with an F1-score of 0.39.

TABLE 9
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 2.4

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.56	0.68	0.61
STRAIGHT HAIR	0.43	0.36	0.39
WAVY HAIR	0.56	0.55	0.55

The confusion matrix in Fig. 20 shows that curly and wavy hair were increasingly well classified (38 and 40 correct predictions, respectively). However, straight hair continued to pose challenges, with 26 of 67 samples misclassified as wavy and 17 as curly. This persistent confusion suggests that additional filters helped with some classes, but feature separability remained limited for more subtle patterns.

The 4-6-8 filter configuration offered moderate gains over the previous lightweight model, particularly in the classification of curly and wavy hair. The increased filter depth allowed the network to learn a richer set of features. However, the model still struggled with straight hair, indicating that further increases in filter complexity, or targeted enhancements like feature normalization or residual learning may be necessary to capture lower-contrast hair types.

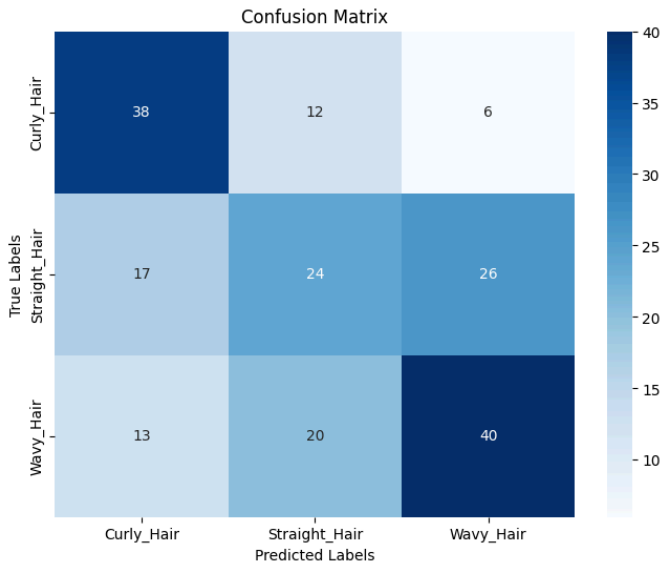


Fig 20. Confusion Matrix for Experiment 3.2

3. FILTERS 8, 16, 32

This sub-experiment examined the impact of increasing convolutional filter counts to 8, 16, and 32 across successive layers. The intent was to assess whether scaling up filter depth would improve the model's capacity to extract hierarchical features and reduce class overlap, particularly for the previously underperforming straight hair category.

As illustrated in Fig. 21, the model demonstrated consistent learning during training, with accuracy reaching 79.31% and loss decreasing to 0.5080. Validation accuracy stabilized at 55.61%, while validation loss hovered around 1.08. Although the training accuracy steadily improved, the validation performance plateaued early, showing mild signs of overfitting.

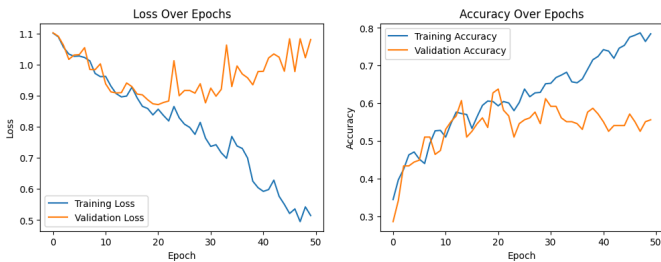


Fig. 21. Accuracy and Loss Functions over Epochs for Experiment 3.3

The classification report in Table XII highlights notable gains in class-level performance. Wavy hair achieved the highest F1-score at 0.61, followed

closely by curly hair at 0.60. However, straight hair performance remained limited with an F1-score of 0.42, though its precision was surprisingly high at 0.64, suggesting confident but narrowly scoped predictions.

TABLE 10
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 3.3

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.55	0.64	0.60
STRAIGHT HAIR	0.64	0.31	0.42
WAVY HAIR	0.53	0.71	0.61

The confusion matrix in Fig. 22 shows that wavy hair classification improved significantly (52 correct). Curly hair remained stable with 36 correctly identified samples. Straight hair, while receiving higher precision, suffered from poor recall, with 30 out of 67 samples misclassified as wavy and 16 as curly.

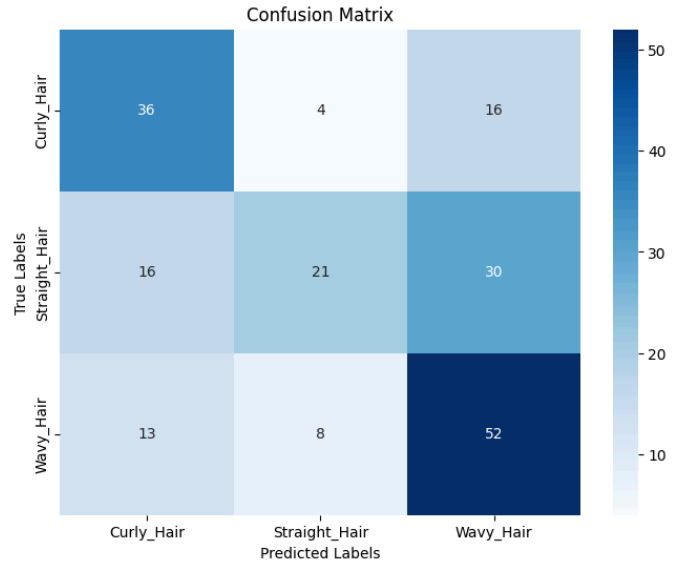


Fig 22. Confusion Matrix for Experiment 3.3

Increasing the number of filters to 8, 16, and 32 enabled the model to better extract and distinguish fine-grained features, leading to improved classification, particularly for wavy hair. Curly hair remained consistently well classified. However, straight hair continued to underperform in recall despite higher precision, implying that while the model could detect straight hair features when present, it often failed to generalize them. This

highlights that further architectural adjustments or specialized features may be required to robustly model linear, low-texture patterns.

4. FILTERS 16, 32, 64

This sub-experiment evaluated a deeper convolutional network with filter sizes 16, 32, and 64, respectively. The objective was to determine whether increasing model capacity beyond the previous setup (8–16–32) would yield further improvements in learning fine-grained visual patterns while minimizing class ambiguity.

As shown in Fig. 23, training accuracy reached a high of 88.34% with a significant reduction in training loss to 0.2871. However, the validation accuracy only reached 58.67%, and validation loss climbed to 1.3862, indicating a noticeable overfitting trend. The widening gap between training and validation performance toward later epochs confirms that the model began to memorize rather than generalize.



Fig. 23. Accuracy and Loss Functions over Epochs for Experiment 3.4

The classification report in Table XI reflects mixed results. Wavy and curly hair achieved high F1-scores of 0.62 and 0.66, respectively, showing clear benefit from the deeper filter architecture. Straight hair performance slightly improved in recall (0.40) and F1-score (0.47), but still lagged behind the other two classes. Overall accuracy reached 59%, the highest so far in this experiment series.

TABLE 11
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 3.4

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.63	0.68	0.66
STRAIGHT HAIR	0.56	0.40	0.47
WAVY HAIR	0.57	0.68	0.62

In Fig. 24, curly and wavy hair continued to be well classified (38 and 50 correct, respectively). Straight hair performance, while improved from earlier setups, still showed high confusion; 27 instances each misclassified as wavy and correctly identified, suggesting partial pattern overlap.

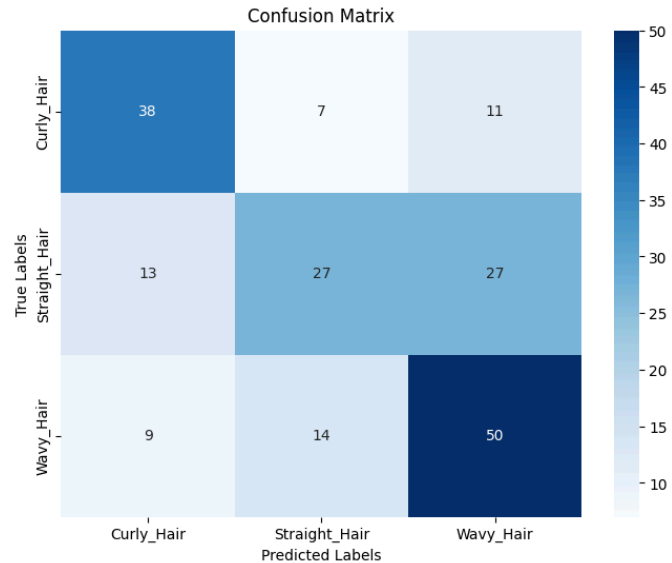


Fig 24. Confusion Matrix for Experiment 3.4

This experiment demonstrated that a deeper model architecture enhances the model’s ability to extract discriminative features, particularly for more textured classes like curly and wavy hair. Straight hair performance improved, but only marginally. The model’s strong training performance alongside a much flatter validation trend clearly indicates overfitting. This suggests the need for additional regularization strategies (e.g., dropout, L2 penalties) or a more diverse dataset to effectively utilize the expanded filter capacity.

5. FILTERS 32, 64, 128

This sub-experiment scaled up the convolutional filter sizes to 32, 64, and 128 across three layers, representing the most computationally intensive model in the filter variation series. The aim was to determine whether deeper representations significantly improve classification performance or merely exacerbate overfitting.

The training progression shown in Fig. 25 reveals a steep and sustained increase in accuracy, reaching 83.44%, with a corresponding drop in loss to 0.4339. However, validation metrics plateaued

early, with final validation accuracy at 55.10% and a higher validation loss of 1.4131, indicating more pronounced overfitting compared to prior experiments.

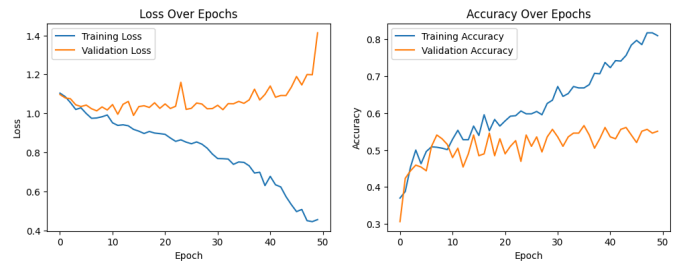


Fig. 25. Accuracy and Loss Functions over Epochs for Experiment 3.5

As seen in Table XII, curly hair classification benefited from this configuration, achieving an F1-score of 0.62 and recall of 0.73, the highest among all sub-experiments so far. Wavy hair followed with an F1-score of 0.56, while straight hair remained problematic with an F1-score of 0.47. The model’s overall accuracy was 55%, on par with prior experiments despite its increased complexity.

TABLE 12
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 3.5

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.53	0.73	0.62
STRAIGHT HAIR	0.54	0.42	0.47
WAVY HAIR	0.58	0.53	0.56

The confusion matrix in Fig. 26 shows a strong improvement in curly hair classification (41 correct). However, confusion persisted for straight and wavy hair classes. Only 28 straight hair samples were correctly classified, with 20 and 19 misclassified as curly and wavy, respectively. This pattern mirrors earlier experiments, reinforcing the observation that straight hair features remain harder to model.

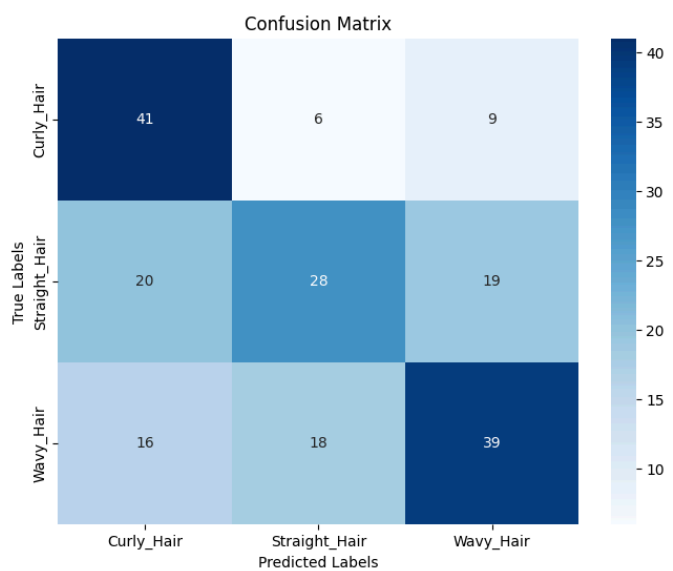


Fig 26. Confusion Matrix for Experiment 3.5

While expanding to 32–64–128 filters improved the network’s expressive power, it did not yield proportional gains in classification performance. In fact, the overfitting observed was more pronounced, evidenced by the diverging training-validation loss and stagnant validation accuracy. These results suggest diminishing returns with filter depth beyond a certain threshold, especially in the absence of additional regularization or data augmentation techniques.

6. FILTERS 16, 16, 16

The final sub-experiment in this filter variation series tested a symmetric filter configuration of three convolutional layers each with 16 filters. The intention was to evaluate whether uniform capacity across layers could provide a balanced trade-off between feature learning and overfitting mitigation.

As illustrated in Fig. 27, the model reached a peak training accuracy of 72.6% while the validation accuracy plateaued around 53.5%. The gap between the training and validation accuracy curves was smaller than in previous configurations with more aggressive filter scaling, indicating moderate overfitting but improved generalization relative to more complex models.



Fig. 27. Accuracy and Loss Functions over Epochs for Experiment 3.6

The classification report (Table XIII) reveals that Curly Hair remained the best classified class with an F1-score of 0.61, while both Straight and Wavy Hair followed closely behind with F1-scores of 0.49 and 0.51 respectively. The balanced filter design helped maintain moderate classification performance across all categories.

TABLE 13
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 3.6

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.53	0.71	0.61
STRAIGHT HAIR	0.52	0.46	0.49
WAVY HAIR	0.57	0.47	0.51

As visualized in Fig. 28, misclassifications among Straight and Wavy Hair classes remained common, reflecting lingering ambiguities in texture pattern recognition. However, the confusion matrix also highlighted a better differentiation of Curly Hair compared to earlier filter configurations, suggesting that moderate filter depth was adequate for capturing its distinct features.

The use of a uniform filter configuration in Experiment 3.6 (16, 16, 16) demonstrated that moderate convolutional capacity can yield stable training dynamics and prevent excessive overfitting. The nearly parallel training and validation curves suggest a more balanced learning process compared to deeper or asymmetrical configurations explored in earlier sub-experiments. Notably, this configuration still enabled meaningful feature extraction, particularly for curly and wavy hair types.

However, the model's performance was still hindered by confusion between straight and wavy hair categories. This implies that while symmetric

filters can promote consistency, they may lack the hierarchical complexity needed for robust separation of visually similar classes. Furthermore, the relatively lower F1-scores for straight hair hint that uniform filters may insufficiently capture finer linear texture patterns. Future work should consider hybrid architectures combining both increasing and uniform filter strategies, or introduce additional layers focused on class-specific feature refinement.

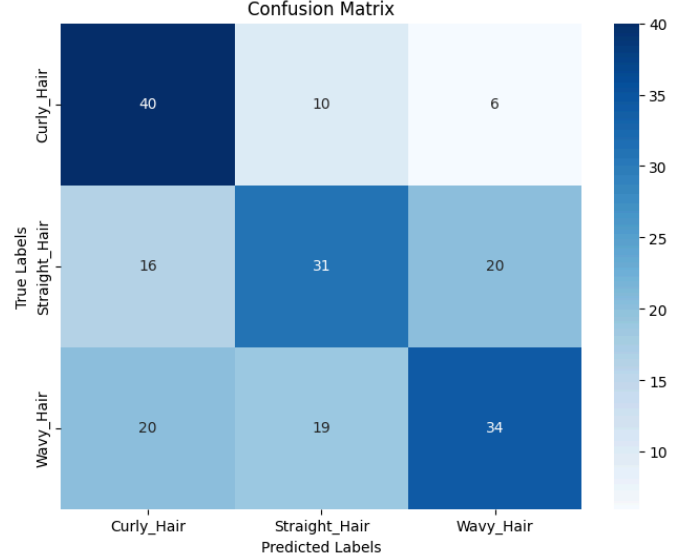


Fig 28. Confusion Matrix for Experiment 3.6

D. Experiment #4: Changing Kernel Size

1. KERNEL SIZE (3,3), (5,5), (9,9)

This sub-experiment tested the effects of progressively increasing kernel sizes across the convolutional layers — starting with (3×3), followed by (5×5), and ending with a larger (9×9) kernel. The hypothesis was that larger receptive fields may enhance the capture of broader texture structures inherent in different hair types, potentially improving classification performance.

As seen in Fig. 29, the training accuracy approached near perfection, reaching 99.42%, while training loss dropped significantly to 0.0263. In contrast, the validation accuracy plateaued around 62.76%, and validation loss remained high at 1.9328, indicating strong signs of overfitting. The divergence in performance began around epoch 10, with validation loss steadily increasing while training continued to improve.

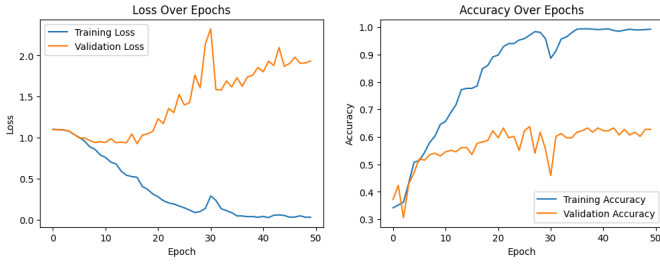


Fig. 29. Accuracy and Loss Functions over Epochs for Experiment 4.1

The classification report (Table X) reveals that curly hair remained the most accurately predicted class with an F1-score of 0.66. Wavy and straight hair followed with F1-scores of 0.64 and 0.58, respectively. While the model demonstrated a reasonably balanced classification across classes, the marginal improvement over simpler configurations did not justify the evident overfitting risk.

TABLE 14
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 4.1

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.63	0.68	0.66
STRAIGHT HAIR	0.60	0.57	0.58
WAVY HAIR	0.64	0.64	0.64

The confusion matrix in Fig. showed that curly hair was consistently well predicted, with 41 correct classifications. However, straight hair still suffered from confusion — being misclassified 17 times as wavy and 14 times as curly. Similarly, 16 wavy hair samples were misclassified as straight. This suggests that, despite broader kernels, inter-class texture ambiguity remained unresolved.

The progressive kernel size approach increased the receptive field across layers, theoretically allowing the model to capture broader spatial features. However, the dramatic overfitting observed suggests that such large kernels (especially the 9×9 in deeper layers) may have led to excessive parameterization relative to the dataset size. This caused the model to fit training data too tightly while failing to generalize to new examples.

Moreover, while performance across all three classes was balanced, the actual improvement in

classification accuracy was marginal compared to less complex models. These findings suggest that kernel size increases should be applied conservatively and ideally paired with regularization mechanisms to mitigate overfitting.

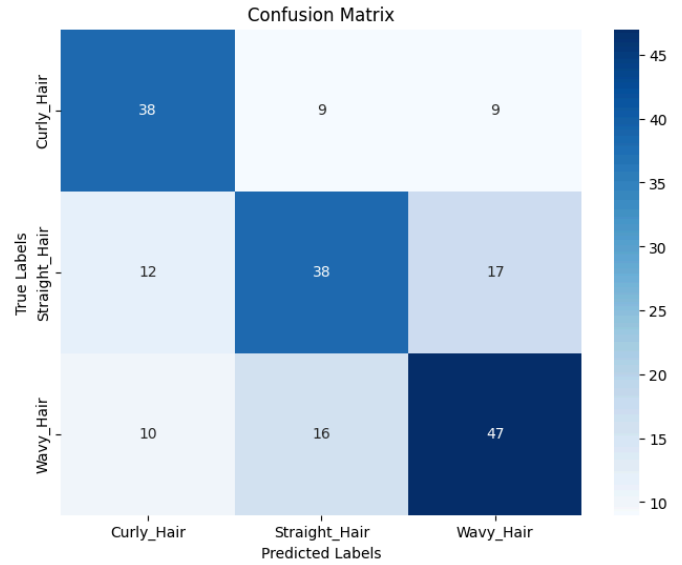


Fig 30. Confusion Matrix for Experiment 4.1

2. KERNEL SIZE (3,3), (3,3), (3,3)

This sub-experiment explored the impact of using a consistent small kernel size of (3,3) across all three convolutional layers. The goal was to observe how local feature extraction with small receptive fields would affect the model's ability to learn discriminative features across the hair types dataset.

As illustrated in Fig. 31, the model achieved a final training accuracy of 86.69%, indicating strong learning on the training set. However, the validation accuracy plateaued at 57.65%, and the validation loss remained relatively high at 1.1240. This discrepancy between training and validation performance suggests the presence of moderate overfitting, despite the seemingly balanced architecture.

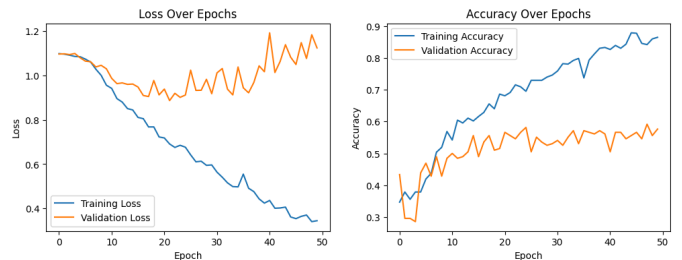


Fig. 31. Accuracy and Loss Functions over Epochs for Experiment 4.2

The classification report in Table 15 highlights varying levels of performance across classes. Wavy hair achieved the highest F1-score at 0.56, while Curly Hair, despite a high precision of 0.60, had a recall of only 0.32, resulting in an F1-score of 0.42. Straight hair showed more balanced precision and recall, yielding an F1-score of 0.50. This indicates the model was able to identify Wavy Hair with better consistency, while Curly Hair was frequently misclassified.

TABLE 15
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 4.2

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.60	0.32	0.42
STRAIGHT HAIR	0.59	0.43	0.50
WAVY HAIR	0.45	0.73	0.56

The confusion matrix in Fig. 32 reveals that Wavy Hair was the most consistently predicted class (53 correct classifications). However, Curly Hair showed high confusion, with 35 samples misclassified as Wavy Hair and only 18 correctly predicted. This imbalance underscores the model's limitations in distinguishing tightly coiled hair textures when relying solely on small receptive fields.

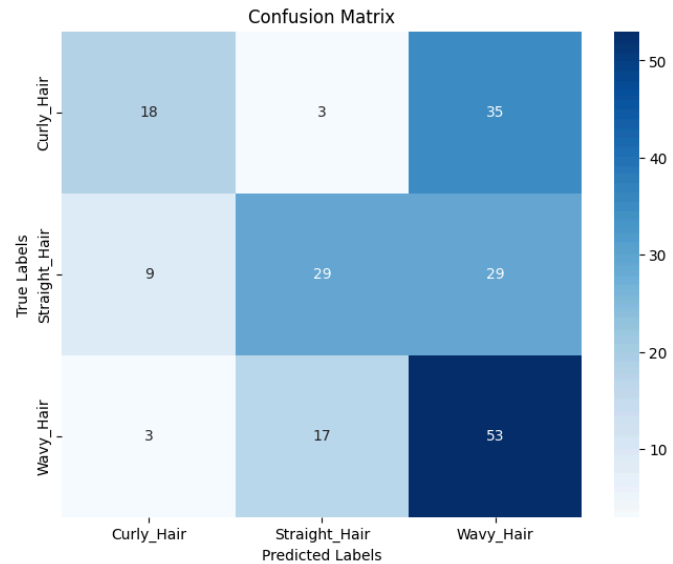


Fig 32. Confusion Matrix for Experiment 4.2

The uniform use of (3,3) kernels allowed the model to focus on fine-grained, localized features. This led to moderate improvements in Wavy Hair classification, but was insufficient to differentiate more complex patterns such as those found in Curly Hair. The sharp drop in recall for Curly Hair suggests that finer spatial patterns were either missed or misinterpreted by the model, leading to frequent misclassification.

Additionally, the observed overfitting indicates that while the model can learn well from training data, it fails to generalize adequately. This could be attributed to the lack of multi-scale feature learning inherent in using uniformly small kernels. Future experiments could benefit from combining small and large kernels or using atrous/dilated convolutions to capture broader spatial context while maintaining resolution.

3. KERNEL SIZE (5,5), (5,5), (5,5)

This sub-experiment examined the effects of using a larger and consistent kernel size of (5,5) across all three convolutional layers. The aim was to assess whether capturing broader spatial features would enhance feature extraction for hair type classification or lead to overfitting due to increased kernel capacity.

As shown in Fig. 33, the model achieved a notably high training accuracy of 98.41%, which reflects strong memorization of the training data. However, the final validation accuracy reached only 65.31%, and the validation loss remained relatively elevated at 1.2276. The large discrepancy between training and validation performance strongly suggests overfitting, as the model struggles to generalize despite its near-perfect training performance.

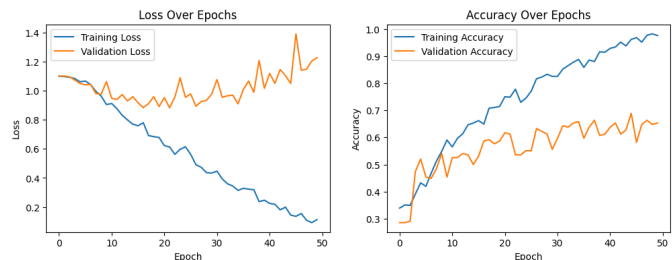


Fig. 33. Accuracy and Loss Functions over Epochs for Experiment 4.3

The classification report in Table XVI indicates fairly balanced performance across the three classes. Wavy Hair achieved the highest recall at 0.73 and an F1-score of 0.67, while Curly Hair also performed reasonably well with an F1-score of 0.68. Straight Hair showed slightly weaker recall (0.54), although its precision (0.68) helped maintain a decent F1-score of 0.60. This suggests that the model is more confident when predicting Wavy and Curly hair, but struggles with consistent identification of Straight Hair.

TABLE 16
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 4.3

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.67	0.70	0.68
STRAIGHT HAIR	0.68	0.54	0.60
WAVY HAIR	0.62	0.73	0.67

The confusion matrix in Fig. 25 further supports this observation. Wavy Hair and Curly Hair were mostly classified correctly (53 and 39 samples, respectively), while Straight Hair exhibited the highest degree of misclassification, with 10 and 21 samples confused as Curly and Wavy Hair, respectively. This suggests that despite the broader receptive field, larger kernels alone may not sufficiently disambiguate linear hair patterns.

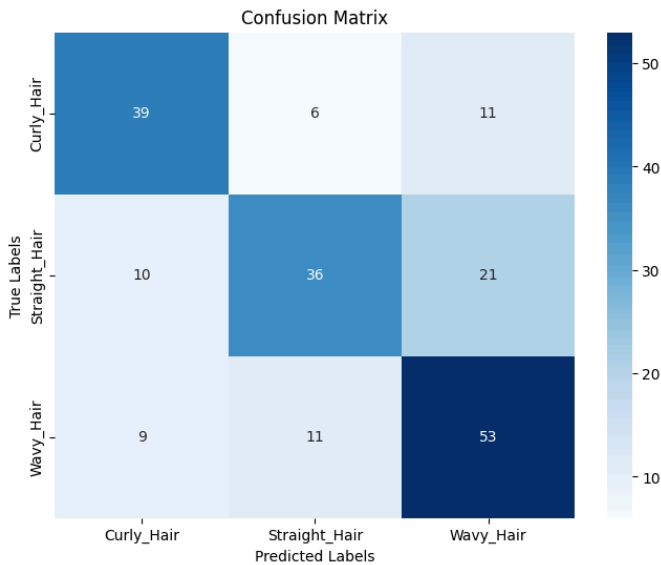


Fig 34. Confusion Matrix for Experiment 4.3

Using a (5,5) kernel size across all layers allowed the model to capture coarser patterns, which slightly improved the performance for Curly and Wavy Hair classes. However, this benefit came at the cost of overfitting. The model's near-perfect training accuracy paired with relatively modest validation performance indicates poor generalization.

Moreover, the over-reliance on large kernels likely reduced the model's sensitivity to finer textures, which are essential for distinguishing between subtle variations in hair structure. The confusion in predicting Straight Hair could be due to the smoothing effect of larger kernels, which may blur out narrow features that distinguish it from Wavy Hair.

Future designs may benefit from hybrid architectures that combine both small and large kernels to balance local and global feature detection. Techniques such as multi-scale convolution or inception modules may offer a better trade-off between expressiveness and generalization.

4. KERNEL SIZE (7,7), (5,5), (3,3)

This sub-experiment investigates the use of decreasing kernel sizes in the CNN architecture, beginning with a large (7,7) kernel in the first convolutional layer and gradually reducing to (5,5) and (3,3) in subsequent layers. The rationale behind this setup is to initially capture broader spatial patterns and then refine the features through progressively smaller receptive fields.

As shown in Fig. 26, the model achieved a training accuracy of 87.56% and a final validation accuracy of 55.61% after 50 epochs. While the training loss steadily decreased to 0.3519, the validation loss remained elevated at 1.2528, showing minimal convergence and indicating overfitting. The widening gap between training and validation metrics illustrates the model's declining generalization ability despite learning from the training data.

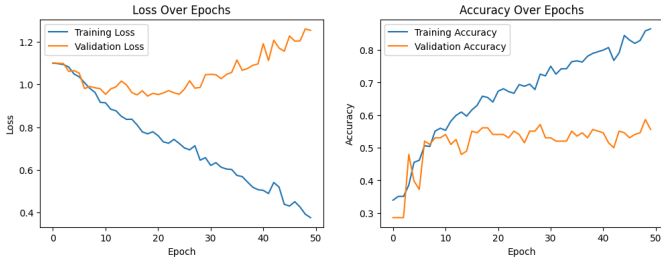


Fig. 35. Accuracy and Loss Functions over Epochs for Experiment 4.4

The classification report in Table XVII presents moderate performance, with Curly Hair achieving the highest F1-score at 0.62. However, Straight Hair was the most problematic, with a notably low recall of 0.36, which negatively impacted its F1-score (0.45). Wavy Hair maintained reasonable performance with an F1-score of 0.58, aided by a recall of 0.62.

TABLE 17
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 4.4

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.54	0.71	0.62
STRAIGHT HAIR	0.62	0.36	0.45
WAVY HAIR	0.54	0.62	0.58

The confusion matrix in Fig. 27 shows that while Curly and Wavy Hair samples were classified relatively accurately (40 and 45 correct predictions, respectively), Straight Hair suffered significant misclassifications—particularly into the Wavy Hair class (27 misclassified samples). This trend highlights persistent confusion between hair types with overlapping textures.

The cascading kernel sizes helped to combine broader contextual feature detection with finer-grained refinement. However, this architecture did not significantly improve validation performance. The model’s overfitting and unstable validation accuracy suggest that, despite structural variety, the model failed to extract generalizable features effectively.

The poor performance on Straight Hair indicates that larger initial kernels may have suppressed critical linear features early in the network, reducing the network’s sensitivity to subtler edge

patterns. Additionally, the overfitting observed implies that while the model captured training patterns well, it lacked sufficient regularization to generalize to unseen samples.

Future improvements may involve experimenting with asymmetric kernels or introducing dropout and batch normalization to mitigate overfitting while retaining multi-scale feature diversity.

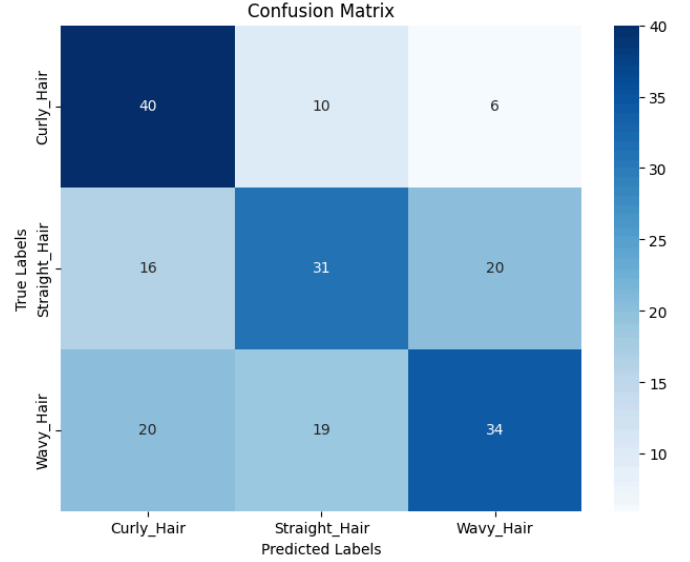


Fig 36. Confusion Matrix for Experiment 4.4

E. Experiment #5: Changing Learning Rate

1. LEARNING RATE 1E-2

This sub-experiment evaluates the impact of using a relatively high learning rate of 1e-2 in training the convolutional neural network. The learning rate is a critical hyperparameter that influences how quickly or smoothly a model converges during training. A higher value can accelerate convergence but may also cause instability or prevent the model from reaching an optimal solution.

As shown in Fig. 37, the model achieved a training accuracy of only 42.32% and a final validation accuracy of 34.69% after 50 epochs. Training loss remained elevated at 1.0588, while validation loss ended at 1.0943, indicating minimal learning. The flatness of both accuracy and loss curves throughout training suggests that the model failed to escape poor initialization or local minima due to the overly aggressive learning rate.

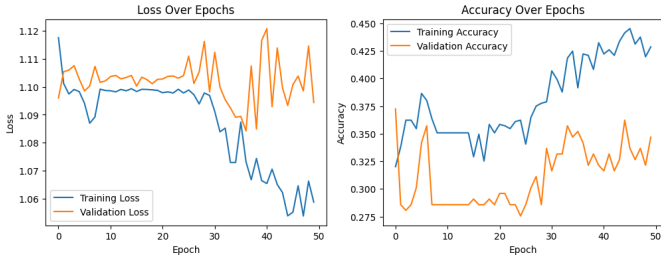


Fig. 37. Accuracy and Loss Functions over Epochs for Experiment 5.1

The classification report in Table XVIII illustrates the model's failure to generalize. Straight Hair was completely unrecognized (0.00 precision, recall, and F1-score), while Wavy Hair and Curly Hair achieved low F1-scores of 0.37 and 0.44, respectively. The model disproportionately favored predicting Curly Hair, regardless of the actual input class.

TABLE 18
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 5.1

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.30	0.82	0.44
STRAIGHT HAIR	0.00	0.00	0.00
WAVY HAIR	0.49	0.30	0.37

The confusion matrix in Fig. 38 confirms this bias, showing almost all predictions defaulted to Curly Hair. Straight Hair was never predicted even once, while a large number of Wavy Hair and Straight Hair samples were misclassified as Curly Hair. This pattern suggests the model converged to a degenerate solution due to the large learning rate.

The results clearly demonstrate that a learning rate of $1e-2$ is too aggressive for this task. The optimizer's large weight updates likely caused erratic behavior or divergence from meaningful learning. This led the model to favor a single dominant class and ignore others completely. This sub-experiment highlights the danger of unstable training dynamics introduced by high learning rates and emphasizes the need for carefully tuned hyperparameters when dealing with sensitive multi-class classification problems.

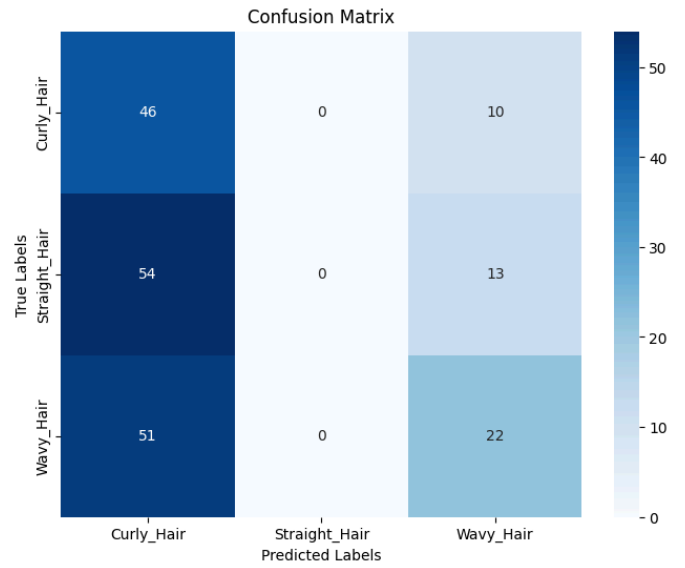


Fig. 38. Confusion Matrix for Experiment 5.1

2. LEARNING RATE $1e-4$

This sub-experiment explores the performance of the CNN model when using a moderate learning rate of $1e-4$, a commonly recommended starting point for many deep learning tasks. This learning rate offers a balance between stability and responsiveness to gradients.

By Epoch 50, the model achieved a training accuracy of 67.02% and a validation accuracy of 55.10%, with respective losses of 0.8244 and 0.9300. As shown in Fig. 39, both the training and validation accuracy steadily improved across the epochs, and the loss curves demonstrate healthy convergence behavior. This indicates that the learning rate allowed the model to learn meaningful patterns without destabilizing training.



Fig. 39. Accuracy and Loss Functions over Epochs for Experiment 5.2

The classification report (Table XIX) shows balanced performance across the three hair types. Curly Hair achieved the highest F1-score of 0.68, while Wavy Hair and Straight Hair followed with

0.53 and 0.43, respectively. The overall performance indicates significant improvement compared to Experiment 5.1.

TABLE 19
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 5.2

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.60	0.79	0.68
STRAIGHT HAIR	0.49	0.39	0.43
WAVY HAIR	0.54	0.52	0.53

The confusion matrix in Fig. 40 reveals that Straight Hair is still relatively difficult to classify, with noticeable confusion with both Curly and Wavy Hair. However, compared to the previous sub-experiment, the model now predicts all classes to some degree of success, reflecting an overall improvement in generalization.

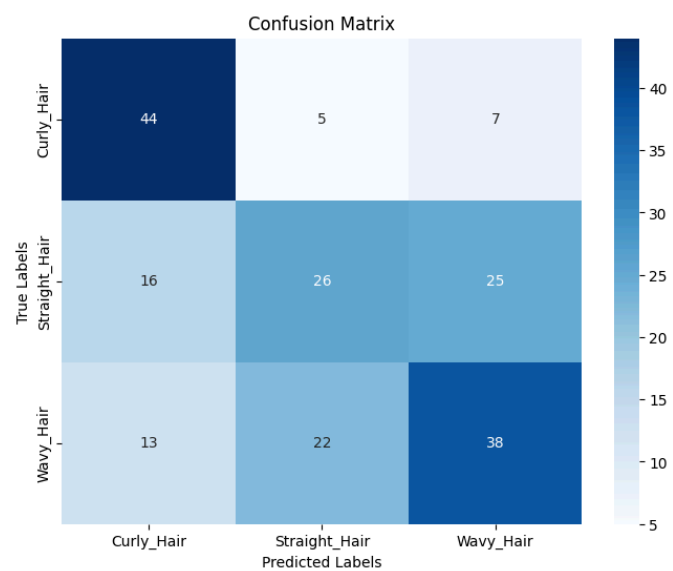


Fig. 40. Confusion Matrix for Experiment 5.2

The confusion matrix in Fig. 40 reveals that Straight Hair is still relatively difficult to classify, with noticeable confusion with both Curly and Wavy Hair. However, compared to the previous sub-experiment, the model now predicts all classes to some degree of success, reflecting an overall improvement in generalization.

3. LEARNING RATE 1e-5

This sub-experiment evaluates the performance of the CNN model when trained with a very small learning rate of 1e-5, intended to examine the effect of slow, incremental weight updates. While this rate can prevent overshooting, it may also hinder convergence if set too low.

After 50 epochs, the model reached a training accuracy of 38.19% and a validation accuracy of 31.63%, with corresponding losses of 1.0837 and 1.0942. As shown in Fig. 41, the loss curves decreased slowly but steadily throughout the training process. However, the accuracy plots reflect a stagnation in validation performance, suggesting underfitting due to insufficient gradient updates.

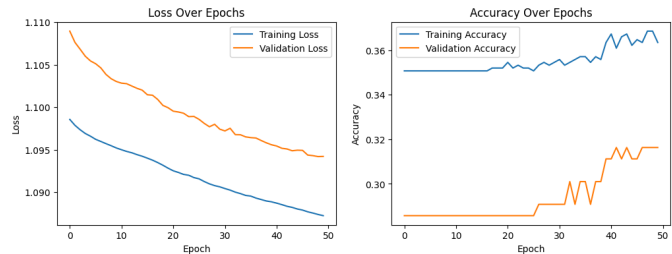


Fig. 41. Accuracy and Loss Functions over Epochs for Experiment 5.3

The classification report in Table XX highlights this limitation clearly. Although Curly Hair achieved a recall of 1.00, the precision was only 0.30, and Wavy Hair received a score of 0.00 across all metrics, indicating the model failed to identify this class altogether. Straight Hair saw similarly poor recall.

TABLE 20
CLASSIFICATION REPORT TABLE FOR EXPERIMENT 5.3

CLASS	PRECISION	RECALL	F1-SCORE
CURLY HAIR	0.30	1.00	0.46
STRAIGHT HAIR	0.67	0.09	0.16
WAVY HAIR	0.00	0.00	0.00

The confusion matrix (Fig. 42) further confirms that the model overwhelmingly favored the Curly Hair class. Most predictions were biased toward this category, with zero correct predictions for Wavy Hair. Such imbalance suggests the model

failed to learn discriminative features for all classes due to inadequate learning progress.

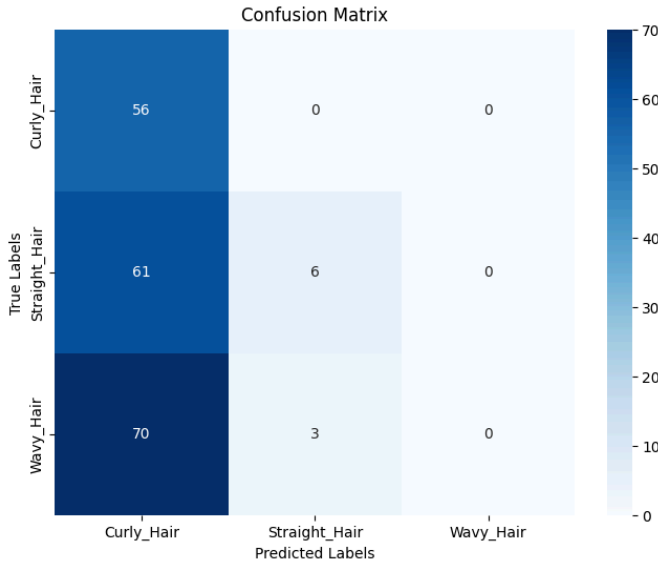


Fig. 42. Confusion Matrix for Experiment 5.3

This result demonstrates that a learning rate of $1e-5$ is too low for this task. The model failed to escape poor local minima or make meaningful updates, leading to extreme class imbalance in predictions and underfitting. Although the loss reduced steadily, the network was unable to generalize or even properly recognize all categories. This experiment underscores the importance of choosing a learning rate that ensures both convergence and feature extraction efficiency.

V. CONCLUSIONS

Among all the CNN models trained, the best model achieved 66% training accuracy and 56% validation accuracy. This means that there is a need to conduct more experiments in order to achieve higher accuracy and better generalization. Overfitting is also an issue in this model since the difference between the training accuracy versus the validation accuracy is significantly large. This shows that the model could not generalize well on unseen data.

It is disappointing that Gaussian Blur with Laplacian Filter dataset made the model perform worse wherein it should help the model perform better because Gaussian Blur smoothens the image and reduces noise while Laplacian Filter sharpens

the edges of the images. One possible way to address this is to lessen the intensity of the Laplacian Filter since some images became unrecognizable. However, the best CNN model performed better on the Gaussian Blur only dataset (based from the “Other Experiment” section in the Jupyter Notebook) achieving 63% training accuracy and 59% validation accuracy, suggesting that the model performs better without the Laplacian Filter applied on the images.

When it comes to the CNN model with the best hyperparameter tuned, the model is effectively learning a relatively simple set of mid-level features from the images in a stable and controlled manner. This shows that the model is relatively compact which can be advantageous in terms of computational cost and memory usage. Additionally, the small learning rate of $1e-4$ contributed to a better generalization compared to other learning rates by preventing the model from overfitting to the training data. Although the model is likely converging to a good solution, it might require more epochs to reach its full potential due to the said small learning rate.

Due to time constraints, there is still a large amount of improvement that can be done in order to further analyze and improve the CNN model. Future experiments should focus on a multi-faceted approach to enhance the CNN model's performance and understanding. Visualizing feature maps, experimenting with data augmentation and regularization techniques, and systematically tuning hyperparameters are crucial steps. Exploring different architectures, including deeper networks and pre-trained models, can also lead to significant improvements. Additionally, techniques like Class Activation Maps and analyzing misclassified images could provide valuable insights into the model's decision-making process, while implementing early stopping can prevent overfitting. By pursuing these experiments, a deeper understanding of the model's behavior can be gained and potentially achieve higher accuracy and better generalization on the hair type classification task.

Overall, this laboratory exercise emphasizes the significance of data preprocessing, hyperparameter tuning, and model architecture modification to deliver the best performance and results of the CNN model.

VI. REFERENCES

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